

RESEARCH ARTICLE

Slip-Compensation-Based Path Tracking Control for Tracked Robots Using VBEKF and Backstepping Control

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ABSTRACT

Tracked robots are widely used in agriculture, military, mining, and other fields. During the traveling process of tracked robots, the complex interaction between the tracks and the ground causes slip, which leads to problems such as low path tracking accuracy and poor control stability. To solve this thorny problem, quantitative control compensation parameters are generally obtained through experiments, or extended Kalman filter (EKF) is designed based on Gaussian noise to estimate slip parameters during motion. However, the sensor measurement noise of tracked robots usually exhibits a non-Gaussian distribution in uneven terrain and variable soil conditions. Under such circumstances, these methods exhibit larger errors and demonstrate inadequate adaptability. Therefore, this paper proposes a variational Bayesian EKF (VBEKF) algorithm for slip parameters estimation, and designs a slip compensation path tracking controller to improve the accuracy and adaptability of the control system of tracked robots under complex operating conditions. The main contributions of this paper are as follows: (1) The non-Gaussian noise was re-modeled using the Student's t-distribution, and combined with variational Bayesian, the VBEKF algorithm was designed. This algorithm can more accurately estimate the slip parameters between the tracks and soil under complex and varying operating conditions, demonstrating enhanced adaptability. (2) Based on the backstepping control principle, a path tracking controller with slip parameter compensation was designed for tracked robots. This controller dynamically adjusts its output control based on the estimated slip parameters to eliminate the impact of slip between the tracks and soil on path tracking accuracy. Finally, the effectiveness of the method was demonstrated through simulations and experiments. This study can improve the adaptability and stability of tracked robots under complex and variable operating conditions, ensuring accurate and rapid task completion, and has broad application prospects.

1 | Introduction

Tracked robots, characterized by their superior trafficability, significant load-bearing capacity, and minimal ground pressure distribution, are widely employed in various scenes, such as agricultural weeding robots, military

transport robots, mining exploration robots, and construction site assistance robots (Zhao et al. 2024; Lu et al. 2023a; Li et al. 2022a). Despite these advantages, in the operating environment of tracked robots, especially in wet and muddy conditions, the strong time-varying slip between the tracks and the ground during operation will reduce their path

tracking accuracy. The ability to achieve precise and stable path tracking control is crucial for the intelligence and practicality of tracked robots (Li et al. 2022b).

Many scholars have conducted research on path tracking control by designing controllers with strong anti-interference capability and robustness or disturbance observation and compensation methods (Li et al. 2021). Kayacan et al. (2018) designed a real-time receding horizon estimation and control framework, capable of achieving good path tracking results even with limited computational resources. Amer et al. (2018) designed an adaptive modified Stanley controller with a fuzzy supervisory system for the path tracking control of tracked armored vehicles. Yang et al. (2021) formulated the vehicle path tracking task as a constrained tracking problem and designed a robust path tracking controller for autonomous vehicles with fuzzy uncertainty. Wang et al. (2021) proposed an improved pure pursuit algorithm for precise and stable path tracking control of tracked pavers. Sabiha et al. (2022) proposed an optimized backstepping controller and an integral sliding mode controller for path tracking control based on the evaluation of the linear and angular velocities required for vehicle kinematics. Lu et al. (2023a) designed a nonlinear disturbance observer to estimate various system disturbances and perform tracking control compensation, which improved the tracking control effect of the vehicle through disturbance compensation. Yu et al. (2021) proposed a receding horizon path tracking control strategy based on a nonlinear disturbance observer, which ensures that the robot asymptotically tracks the desired path in the presence of external disturbances. Zhao et al. (2022) proposed a model predictive multi-objective optimal control method based on extended state observer (ESO), by introducing ESO to estimate in real time the disturbances caused by the unknown external environment and simplified modeling during the vehicle tracking process, and to perform feed-forward compensation to improve the path tracking accuracy. Tan and Zhu (2024) proposed a twin-observer-based path tracking controller to suppress the impact of disturbances or uncertainties on vehicle path tracking performance.

On the other hand, some scholars have focused on estimating the slip parameters, which account for the largest proportion of external disturbances, and performing path prediction or slip compensation control (Nourizadeh et al. 2023; Chen et al. 2023). Ha et al. (2013) propose a new dual estimation algorithm based on the extended Kalman filter (EKF) to estimate the robot's position and wheel slip. Wang et al. (2018) described slip by introducing the instantaneous center of rotation (ICR) into the kinematic model of a tracked robot, using an innovative EKF based on data fusion of encoder and GPS signals and real-time estimation of slip parameters. Zarei et al. (2024) proposed an explicit multi-trajectory fusion algorithm for multi-sensor networks based on unscented Kalman filter (UKF), which can be used for real-time slip parameters estimation and trajectory prediction. Zhao et al. (2024) proposed an improve sliding mode observer (ISMO) method for slip parameters estimation and trajectory prediction, providing theoretical guidance for trajectory planning and tracking control of tracked robots. Xiong and Lu (2017) established a

kinematic model of tracked vehicles based on the instantaneous center of rotation and used the Levenberg-Marquardt (L-M) algorithm to iteratively solve the slip parameters to predict the vehicle trajectory, which reduces the error by 30% compared to the traditional method. Zhao et al. (2019) combined the slip kinematics model with the path tracking control strategy of slip guided tracked vehicles and proposed an MPC that integrates the estimation results of slip parameters to achieve optimal control input for tracked vehicles, which increased the path tracking accuracy by 34.4%. Kang et al. (2022) proposed an improved model that considers the slip of tracked vehicles, and designed a path tracking controller based on deep reinforcement learning. Li et al. (2023) proposed a model predictive control (MPC) based strategy that considers slip, improves the immunity of the trajectory during traveling and reduces the lateral error by 72.5%.

The slip phenomenon between the tracked robots and the ground will cause the actual control amount to be smaller than the expected control amount, thus reducing accuracy of the path tracking control. If the missing control amount caused by the slip can be quantitatively compensated, the accuracy of path tracking control can be effectively improved. According to the literature review above, it can be known that there are two main methods to estimate the slip parameters between the tracks and the ground, namely EKF and SMO. Based on previous research (Zhao et al. 2024), we decided to improve the EKF method, which has a simpler model and higher real-time performance. The sensor measurement noise of tracked robots usually exhibits a non-Gaussian distribution in uneven terrain and variable soil conditions. However, most existing methods for estimating slip parameters between tracks and the ground based on EKF assume that the noise is Gaussian distributed, which limits the improvement of slip parameter estimation performance. For this reason, the non-Gaussian noise was re-modeled using the Student's *t*-distribution, and combined with variational Bayesian, the variational Bayesian EKF (VB-EKF) algorithm was designed to accurately and quickly estimate the slip parameters. Furthermore, based on the backstepping control principle, a path tracking controller with slip parameter compensation was designed for tracked robots. This significantly improves the path tracking accuracy of tracked robots under complex and variable working conditions.

The organization of this paper is as follows: Section 2 establishes the kinematic model of the tracked robot, introduces the method of slip parameter estimation, and designs a path tracking controller with slip compensation. Section 3 conducts simulations and experiments on slip parameter estimation and path tracking control. Finally, Section 4 provides a brief conclusion.

2 | Methodology

2.1 | Kinematic Modeling of Tracked Robot

As depicted in Figure 1, the positive direction of the *X* axis is east, the positive direction of the *y*-axis is north, and *xoy* is the

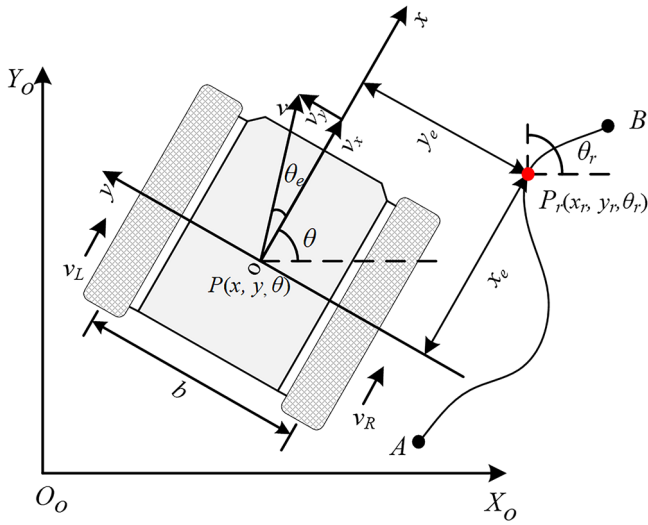


FIGURE 1 | Planar motion diagram of the tracked robot.

local coordinate system of the tracked robot. The origin o coincides with the center of mass of the tracked robot, and the positive direction of the x -axis is the forward direction of the tracked robot. To establish a kinematic model for a tracked robot, the following assumptions are made (Lu et al. 2023b): (1) The length of the tracked robot's tracks is fixed and the ground-contacting parts do not deform; (2) The ground pressure of the track is uniform and the tracked robot moves on a flat ground; (3) The geometric center of the tracked robot coincides with the center of mass.

Based on the above assumptions, it can be obtained (Lu et al. 2023b):

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ -1/b & 1/b \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix}, \quad (1)$$

where b is the center distance of the track, m; r is the radius of the driving wheel; θ is the heading angle of the tracked robot, rad; v_L and v_R represent the theoretical rotational speeds of the left and right tracks, rad/s; v and ω represent the speed and angular velocity of the tracked robot, m/s and rad/s, respectively.

Neglecting the slip (slippage and slip-rotation) between the tracks and the ground during the motion of the tracked robot, the kinematic equations in the global XOY coordinate system can be represented as:

$$\begin{bmatrix} \dot{X}_O \\ \dot{Y}_O \\ \dot{\theta}_O \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} 0.5 \cos \theta & 0.5 \cos \theta \\ 0.5 \sin \theta & 0.5 \sin \theta \\ -1/b & 1/b \end{bmatrix} \begin{bmatrix} v_L \\ v_R \end{bmatrix}. \quad (2)$$

However, in actual motion control, the tracked robot may slip (slippages and slip-rotation) in the forward and lateral directions, resulting in a discrepancy between the output control quantity and the desired control quantity, which in turn causes

error in path tracking control. The slip parameters $S(i_L, i_R, \varphi)$ can be represented as:

$$\begin{cases} i_L = \frac{v_L - v_x}{v_L} \\ i_R = \frac{v_R - v_x}{v_R} \\ \varphi = \arctan \frac{v_y}{v_x} \end{cases}, \quad (3)$$

where v_x and v_y are the speeds of the tracked robot in the x -axis and y -axis directions, respectively, i_L and i_R are the slip rate of the tracked robot in the left and right track traveling directions, and φ is the lateral slip-rotation angle of the tracked robot.

If the influence of slip (slippages and slip-rotation) between the robot's track and the ground is considered, the kinematic equation of the tracked robot in the local xoy coordinate system can be represented as (Lu et al. 2020):

$$\begin{cases} \dot{x} = \frac{1}{2} [v_L(1 - i_L) + v_R(1 - i_R)] = v_x \\ \dot{y} = v_x \tan \varphi = \frac{1}{2} [v_L(1 - i_L) + v_R(1 - i_R)] \tan \varphi \\ \dot{\theta} = \frac{1}{b} [-v_L(1 - i_L) + v_R(1 - i_R)] \end{cases} \quad (4)$$

Therefore, the kinematic model of the tracked robot in the global XOY coordinate system can be obtained through coordinate transformation:

$$\begin{cases} \dot{X}_O = v_x \cos \theta - v_x \tan \varphi \sin \theta \\ \dot{Y}_O = v_x \sin \theta + v_x \tan \varphi \cos \theta \\ \dot{\theta}_O = [2v_x - 2v_L(1 - i_L)]/b = [-2v_x + 2v_R(1 - i_R)]/b \end{cases}. \quad (5)$$

Combining Equations (4) and (5), the kinematic model of the tracked robot in the global XOY coordinate system considering the slip between the tracks and the ground can be rewritten as:

$$\begin{bmatrix} \dot{X}_O \\ \dot{Y}_O \\ \dot{\theta}_O \end{bmatrix} = \begin{bmatrix} 0.5(\cos \theta - \tan \varphi \sin \theta) & 0.5(\cos \theta - \tan \varphi \sin \theta) \\ 0.5(\sin \theta + \tan \varphi \cos \theta) & 0.5(\sin \theta + \tan \varphi \cos \theta) \\ -1/b & 1/b \end{bmatrix} \begin{bmatrix} v_L(1 - i_L) \\ v_R(1 - i_R) \end{bmatrix}. \quad (6)$$

2.2 | Slip Parameter Estimation

2.2.1 | Traditional EKF

According to the kinematic equation depicted in Equation (6), the following state variables are defined $\mathbf{X} = [X_O, Y_O, \theta_O, i_L, i_R, \varphi]^T$, and the system model can be represented by a nonlinear differential equation:

$$\dot{\mathbf{X}} = \mathbf{f}(\mathbf{X}_O, \mathbf{u}, \mathbf{w}), \quad (7)$$

where $\mathbf{u}$ is the control input vector, and $\mathbf{w}$ is the system noise. From Equation (7), it can be derived that (Zhao et al. 2024):

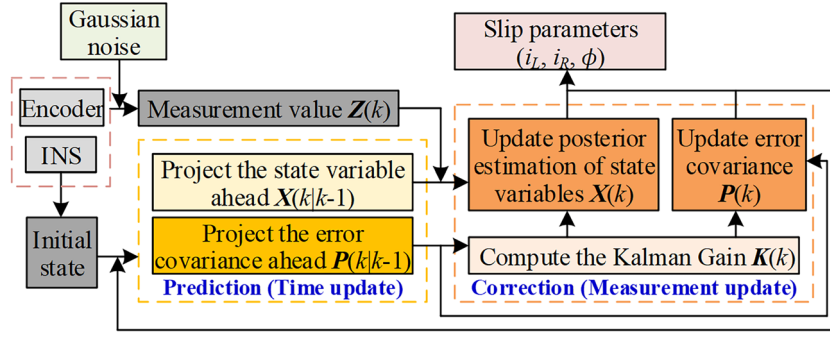


FIGURE 2 | Schematic diagram of the traditional EKF for estimating slip parameters.

$$\dot{\mathbf{X}} = \begin{bmatrix} \dot{X}_O \\ \dot{Y}_O \\ \dot{\theta}_O \\ i_L \\ i_R \\ \phi \end{bmatrix} = \begin{bmatrix} 0.5r\omega_L(\cos\theta - \tan\varphi\sin\theta)(1 - i_L) + 0.5r\omega_R \\ (\cos\theta - \tan\varphi\sin\theta)(1 - i_R) + w_X \\ 0.5r\omega_L(\sin\theta + \tan\varphi\cos\theta)(1 - i_L) + 0.5r\omega_R \\ (\sin\theta + \tan\varphi\cos\theta)(1 - i_R) + w_Y \\ -r\omega_L(1 - i_L)/b + r\omega_R(1 - i_R)/b + w_\theta \\ w_L \\ w_R \\ w_\varphi \end{bmatrix}, \quad (8)$$

where w_X , w_Y , w_θ , w_L , w_R and w_φ are zero-mean process noises.

Discrete Equation (8) and we set the sampling time interval of the EKF as Δt . The covariance of the EKF prediction can be represented as:

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k-1}\mathbf{P}_{k-1}\mathbf{F}_{k-1}^T + \mathbf{L}_{k-1}\mathbf{Q}_{k-1}\mathbf{L}_{k-1}^T. \quad (9)$$

Among them, $\mathbf{Q}$ is the process noise covariance matrix, $\mathbf{F}_{k-1}$ and $\mathbf{L}_{k-1}$ are the Jacobian matrix, which are represented as follows (Wu et al. 2015):

$$\mathbf{F}_{k-1} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{X}} \right|_{\mathbf{X}_{k-1}} = \begin{bmatrix} \mathbf{F}_p & \mathbf{F}_R \\ \mathbf{O}_3 & \mathbf{I}_3 \end{bmatrix} \Delta t, \quad (10)$$

$$\mathbf{L}_{k-1} = \left. \frac{\partial \mathbf{f}}{\partial \mathbf{w}} \right|_{\mathbf{X}_{k-1}} = \mathbf{I}_6 \Delta t, \quad (11)$$

where $\mathbf{I}_n$ and $\mathbf{O}_n$, $n \in \mathbf{N}$ represent $n \times n$ identity matrix and zero matrix, respectively; $\mathbf{F}_p = \begin{bmatrix} 1 & 0 & -\dot{X}_O \\ 0 & 1 & \dot{Y}_O \\ 0 & 0 & 1 \end{bmatrix}$,

$$\mathbf{F}_R = \begin{bmatrix} -0.5v_L(\cos\theta - \sin\theta\tan\varphi) & -0.5v_R(\cos\theta - \sin\theta\tan\varphi) & 0.5\sin\theta(\tan^2\varphi + 1)(v_L(i_L - 1) + v_R(i_R - 1)) \\ -0.5v_L(\sin\theta + \cos\theta\tan\varphi) & -0.5v_R(\sin\theta + \cos\theta\tan\varphi) & -0.5\cos\theta(\tan^2\varphi + 1)(v_L(i_L - 1) + v_R(i_R - 1)) \\ v_L/b & -v_R/b & 0 \end{bmatrix}.$$

The position information of the tracked robot in the global coordinate system obtained by the integrated navigation system (INS) is as follows (Ha et al. 2013; Chen et al. 2019):

$$\mathbf{Z}_k = \begin{bmatrix} X_k \\ Y_k \\ \theta_k \end{bmatrix} = \mathbf{H}_k \mathbf{X}_k + \mathbf{V}_k, \quad (12)$$

$$\text{where the measurement matrix } \mathbf{H}_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

The Kalman gain coefficient can be represented as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T [\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k]^{-1}. \quad (13)$$

The state and covariance estimates are updated as follows:

$$\mathbf{X}_k = \mathbf{X}_{k|k-1} + \mathbf{K}_k [\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_{k|k-1}], \quad (14)$$

$$\mathbf{P}_k = [\mathbf{I} - \mathbf{K}_k \mathbf{H}_k] \mathbf{P}_{k|k-1}, \quad (15)$$

The principle of estimating the slip parameters between the tracks and the ground for a tracked robot based on the traditional EKF method is illustrated in Figure 2 (Zhao et al. 2024).

2.2.2 | Proposed VBEKF

The EKF algorithm can accurately estimate the state and predict the parameters of nonlinear systems only when the structural parameters of the dynamic system are known and the process noise and observation noise strictly follow Gaussian distributions. However, when a tracked robot operates in a complex environment, the soil conditions and terrain randomly change, and the

noise exhibits heavy-tailed characteristics, including extreme values that do not follow Gaussian distributions. In this case, the accuracy of the EKF algorithm significantly decreases. As depicted in Figure 3, the Student's t-distribution has excellent

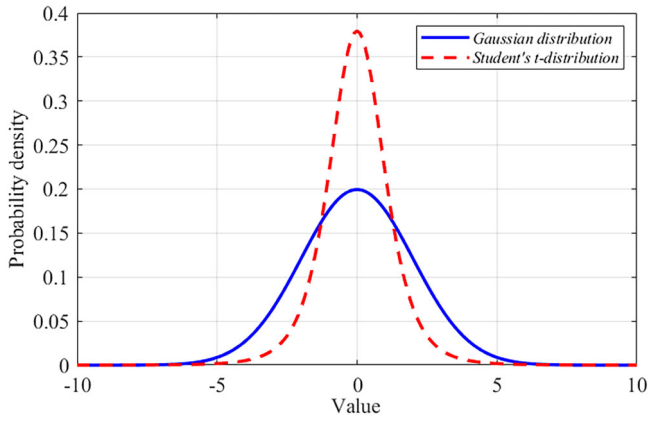


FIGURE 3 | Gaussian distribution and Student's t -distribution.

heavy-tailed characteristics and can better simulate the presence of abnormal extreme values. Therefore, to better accommodate the field environment, this paper employs the Student's t -distribution for noise reconstruction, and devised a slip parameters estimator, which seamlessly integrates variational Bayesian techniques with the EKF. The proposed VBEKF algorithm has better accuracy and robustness in the face of complex environments and can effectively improve the navigation and control accuracy of tracked robots.

The heavy-tailed noise is modelled as follows (Chen et al. 2019):

$$p(\mathbf{v}_k) = St(\mathbf{v}_k; \mathbf{0}, \mathbf{R}_k, \nu), \quad (16)$$

where the parameters are the mean vector $\mathbf{0}$, the scale matrix $\mathbf{R}_k$ (nominal covariance matrix of measurement noise), and the degree of freedom parameter ν , respectively. The likelihood probability density function is constructed based on Equations (12) and (16):

$$p(\mathbf{Z}_k|\mathbf{X}_k) = St(\mathbf{Z}_k; \mathbf{H}_k\mathbf{X}_k, \mathbf{R}_k, \nu), \quad (17)$$

where $\mathbf{H}_k$ is the partial derivative Jacobian matrix of $\mathbf{H}$ with respect to $\mathbf{x}$.

Next, to model the noise of the heavy-tailed process, the one-step prediction probability density function $p(\mathbf{X}_k|\mathbf{Z}_{1:k-1})$ is approximated as the Student's t -distribution:

$$p(\mathbf{X}_k|\mathbf{Z}_{1:k-1}) = St(\mathbf{X}_k; \mu_k, \Sigma_k, \omega), \quad (18)$$

where μ_k is the mean vector, Σ_k is the scale matrix, and ω is the degree of freedom parameter. According to Bayes' theorem, the one-step prediction probability density function $p(\mathbf{X}_k|\mathbf{Z}_{1:k-1})$ is represented as (Xu et al. 2013):

$$\begin{aligned} p(\mathbf{X}_k|\mathbf{Z}_{1:k-1}) &= \int p(\mathbf{X}_k|\mathbf{X}_{k-1})p(\mathbf{X}_{k-1}|\mathbf{Z}_{1:k-1})d\mathbf{X}_{k-1} \\ &= \int p_w(\mathbf{X}_k - \mathbf{F}_k\mathbf{X}_{k-1})p(\mathbf{X}_{k-1}|\mathbf{Z}_{1:k-1})d\mathbf{X}_{k-1} \end{aligned} \quad (19)$$

where p_w is the probability density function of process noise.

Considering that the process noise vector $\mathbf{w}_k$ and the nominal covariance matrix $\mathbf{Q}_k$ have a mean value of zero, according to Equation (19), the calculation for the mean vector μ_k and the nominal covariance matrix $\Delta_{k|k-1}$ is represented as:

$$\mu_k = \mathbf{F}_{k-1}\hat{\mathbf{X}}_{k-1|k-1} = \hat{\mathbf{X}}_{k-1}, \quad (20)$$

$$\Delta_{k|k-1} = \mathbf{F}_k\mathbf{P}_{k-1|k-1}\mathbf{F}_k^T + \mathbf{Q}_k = \mathbf{P}_{k|k-1}. \quad (21)$$

The nominal covariance matrix $\mathbf{P}_{k|k-1}$ can be used for one-step prediction of the scale matrix Σ_k of the probability density function $p(\mathbf{X}_k|\mathbf{Z}_{1:k-1})$. However, in the actual slip parameter estimation process, $\mathbf{Q}_{k-1}$ is difficult to be obtained directly from measurements, and the heavy-tailed process noise leads to the uncertainty of the scale matrix Σ_k to increase, which makes the performance of the slip parameter estimation decrease. Therefore, the variational Bayesian algorithm can be used to estimate the scale matrix Σ_k , and the inverse Wishart distribution is chosen as the conjugate prior of the covariance matrix when the mean of the Gaussian distribution is unknown.

Since the Student's t -distribution is not completely closed, the posterior probability density function cannot be computed directly. Two auxiliary random variables can be introduced to transform the state-space model based on the Student's t -distribution into a hierarchical Gaussian state-space model. The predicted probability density $p(\mathbf{X}_k|\mathbf{Z}_{1:k-1})$ and the likelihood probability density $p(\mathbf{Z}_k|\mathbf{X}_k)$ can be rewritten as:

$$p(\mathbf{X}_k|\mathbf{Z}_{1:k-1}, \xi_k) = \int N(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k-1}, \Sigma_k/\xi_k)G\left(\xi_k; \frac{\omega}{2}, \frac{\omega}{2}\right)d\xi_k, \quad (22)$$

$$p(\mathbf{Z}_k|\mathbf{X}_k) = \int N(\mathbf{Z}_k; \mathbf{H}_k\mathbf{X}_k, \mathbf{R}_k/\lambda_k)G\left(\lambda_k; \frac{\nu}{2}, \frac{\nu}{2}\right)d\lambda_k, \quad (23)$$

where $G(\cdot; \alpha, \beta)$ is the gamma probability density function with shape parameter α and rate parameter β , and ξ_k and λ_k are auxiliary random variables. According to Equations (22) and (23), the one-step prediction probability density and the likelihood probability density can be rewritten in the following hierarchical Gaussian form:

$$\begin{aligned} p(\mathbf{X}_k|\xi_k, \mathbf{Z}_{1:k-1}) &= N(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k-1}, \Sigma_k/\xi_k) \\ p(\xi_k) &= G\left(\xi_k; \frac{\omega}{2}, \frac{\omega}{2}\right) \end{aligned} \quad (24)$$

$$\begin{aligned} p(\mathbf{Z}_k|\mathbf{X}_k, \lambda_k) &= N(\mathbf{Z}_k; \mathbf{H}_k\mathbf{X}_k, \mathbf{R}_k/\lambda_k) \\ p(\lambda_k) &= G\left(\lambda_k; \frac{\nu}{2}, \frac{\nu}{2}\right) \end{aligned} \quad (25)$$

Equations (20), (21), (24), and (25) constitute a hierarchical Gaussian state-space model based on the Student's t -distribution, so that the state estimation problem for a linear state-space model with a heavy-tailed process and measurement noise can be transformed into a state estimation problem for a hierarchical Gaussian state-space model based on the Student's t -distribution. To estimate the state $\mathbf{X}_k$ in a hierarchical

Gaussian state space model, the joint posterior probability density function $p(\mathbf{X}_k, \xi_k, \Sigma_k, \lambda_k | \mathbf{Z}_{1:k})$ needs to be computed. For the spatial model of the hierarchical Gaussian state, it is not possible to compute the a posteriori probability density function directly, so the posterior probability density function is approximated using a variational Bayesian based approach to solve for:

$$p(\mathbf{X}_k, \xi_k, \Sigma_k, \lambda_k | \mathbf{Z}_{1:k}) \approx q(\mathbf{X}_k)q(\xi_k)q(\Sigma_k)q(\lambda_k), \quad (26)$$

where $q(\cdot)$ is the approximate posterior probability density function. According to the standard variational Bayesian algorithm, the approximate posterior probability density function can be obtained by minimizing the Kullback-Leibler divergence between the approximate posterior probability density function $\{q(\mathbf{X}_k), q(\xi_k), q(\Sigma_k), q(\lambda_k)\}$ and the true posterior probability density function $p(\mathbf{X}_k, \xi_k, \Sigma_k, \lambda_k | \mathbf{Z}_{1:k})$:

$$\{q(\mathbf{X}_k), q(\xi_k), q(\Sigma_k), q(\lambda_k)\} = \arg \min \text{KLD}(q(\mathbf{X}_k)q(\xi_k) \times q(\Sigma_k)q(\lambda_k) || p(\mathbf{X}_k, \xi_k, \Sigma_k, \lambda_k | \mathbf{Z}_{1:k})), \quad (27)$$

where $\text{KLD}(q(\mathbf{X}) || p(\mathbf{X})) \triangleq \int q(\mathbf{X}) \log \frac{q(\mathbf{X})}{p(\mathbf{X})} d\mathbf{X}$ is the Kullback-Leibler divergence, the optimal solution of Equation (27) should satisfy the following relationship (Chen et al. 2019):

$$\log q(\phi) = E_{\theta^{(-\phi)}}[\log p(\theta, \mathbf{Z}_{1:k})] + c_\phi \quad (28)$$

$$\theta \triangleq \{\mathbf{X}_k, \xi_k, \Sigma_k, \lambda_k\} \quad (29)$$

where ϕ is any element in θ , $\theta^{(-\phi)}$ is the set of θ except ϕ , $E[\cdot]$ is the computational expectation, and c_ϕ is the constant of correlation with variable ϕ . Since $q(\mathbf{X}_k)$, $q(\xi_k)$, $q(\Sigma_k)$ and $q(\lambda_k)$ are coupled with each other and cannot be solved directly in Equation (28). This paper adopts an iterative updating method to gradually approximate the optimal solution by updating a subset of the parameters in each iteration. This method does not provide an analytical solution, but can find a local optimal solution through iteration, where only one factor in Equation (26) is changed and all other factors are kept constant.

1. Calculation of approximate a posteriori probability density function

Based on the hierarchical Gaussian state space model and Bayes' theorem, the joint posteriori probability density function $p(\theta, \mathbf{Z}_{1:k})$ can be decomposed as (Xu et al. 2013):

$$p(\theta | \mathbf{Z}_{1:k}) = p(\mathbf{Z}_k | \mathbf{X}_k, \lambda_k) p(\mathbf{X}_k | \xi_k, \mathbf{Z}_{1:k-1}) p(\xi_k) p(\Sigma_k) p(\lambda_k). \quad (30)$$

By substituting Equations (24) and (25) into Equation (30), it can be obtained:

$$p(\theta | \mathbf{Z}_{1:k}) = N(\mathbf{Z}_k; \mathbf{H}_k \mathbf{X}_k, \frac{\mathbf{R}_k}{\lambda_k}) N(\mathbf{X}_k; \xi_{k|k-1}, \frac{\Sigma_k}{\xi_k}) \times G(\xi_k; \frac{\omega}{2}, \frac{\omega}{2}) \text{IW}(\Sigma_k; u_k, \mathbf{U}_k) G(\lambda_k; \frac{\nu}{2}, \frac{\nu}{2}) \quad (31)$$

It can be further obtained as:

$$\begin{aligned} \log p(\theta | \mathbf{Z}_{1:k}) &= \left(\frac{m+\nu}{2} - 1 \right) \log \\ &\lambda_k - \frac{\nu}{2\lambda_k} - \frac{\lambda_k}{2} (\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)^T \mathbf{R}_k^{-1} (\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k) \\ &+ \left(\frac{n+\omega}{2} - 1 \right) \log \xi_k - \frac{\omega}{2\xi_k} (\mathbf{X}_k - \xi_{k|k-1})^T \Sigma_{k|k-1}^{-1} \cdot \\ &\quad (\mathbf{X}_k - \xi_{k|k-1}) \\ &- \frac{1}{2} \text{tr}(\mathbf{U}_k^T \Sigma_k^{-1}) - \frac{1}{2} (n + u_k + 2) \log |\Sigma_k| \end{aligned} \quad (32)$$

Setting $\Phi = \xi_k$ and bringing Equation (32) into Equation (28), it can be obtained:

$$\begin{aligned} \log q^{(i+1)}(\xi_k) &= \left(\frac{n+\omega}{2} - 1 \right) \\ \log \xi_k - 0.5 \left(\omega + \text{tr}(\mathbf{D}_k^{(i)} E^{(i)}[\Sigma_k^{-1}]) \right) \xi_k + c_\xi, \end{aligned} \quad (33)$$

where $q^{(i+1)}(\cdot)$ is an approximation of the probability density function $q(\cdot)$ for $i+1$ iterations, $E^{(i)}[\cdot]$ is the expected value of the variable p for the i th iteration, $\text{tr}(\cdot)$ is the trace operation, and $\mathbf{D}_k^{(i)}$ is represented as:

$$\mathbf{D}_k^{(i)} = E^{(i)}[(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1})(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1})^T]. \quad (34)$$

According to Equation (33), $q^{(i+1)}(\cdot)$ can be updated to a gamma probability density function containing the shape parameter α_k^{i+1} and the rate parameter β_k^{i+1} , which is represented as:

$$q^{(i+1)}(\xi_k) = G(\xi_k; \alpha_k^{i+1}, \beta_k^{i+1}), \quad (35)$$

where the shape parameter α_k^{i+1} and the rate parameter β_k^{i+1} are defined as:

$$\alpha_k^{i+1} = 0.5(n + \omega), \quad (36)$$

$$\beta_k^{i+1} = 0.5 \left(\omega + \text{tr}(\mathbf{D}_k^{(i)} E^{(i)}[\Sigma_k^{-1}]) \right). \quad (37)$$

Setting $\Phi = \lambda_k$ and bringing Equation (34) into Equation (30), it can be obtained (Xu et al. 2013):

$$\begin{aligned} \log q^{(i+1)}(\lambda_k) &= \left(\frac{m+\nu}{2} - 1 \right) \\ \log \lambda_k - 0.5 \left\{ \nu + \text{tr}(\mathbf{E}_k^{(i)} \mathbf{R}_k^{-1}) \right\} \lambda_k + c_\lambda, \end{aligned} \quad (38)$$

where

$$\mathbf{E}_k^{(i)} = E^{(i)}[(\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)(\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)^T]. \quad (39)$$

According to Equation (38), $q^{(i+1)}(\lambda_k)$ can be updated to a gamma probability density function with a shape parameter γ_k^{i+1} and a rate parameter δ_k^{i+1} :

$$q^{(i+1)}(\lambda_k) = G(\lambda_k; \gamma_k^{i+1}, \delta_k^{i+1}) \quad (40)$$

where the shape parameter γ_k^{i+1} and the rate parameter δ_k^{i+1} are represented as:

$$\gamma_k^{i+1} = 0.5(m + \nu) \quad (41)$$

$$\delta_k^{i+1} = 0.5\left(\nu + \text{tr}\left(\mathbf{E}_k^{(i)} \mathbf{R}_k^{-1}\right)\right) \quad (42)$$

Setting $\Phi = \Sigma_k$ and bringing Equation (32) into Equation (28), it can be obtained (Xu et al. 2013):

$$\begin{aligned} \log q^{(i+1)}(\Sigma_k) &= -\frac{1}{2}(n + u_k + 2)\log|\Sigma_k| \\ &\quad - \frac{1}{2}\text{tr}\left(\mathbf{U}_k + \mathbf{E}^{(i+1)}\left[\xi_k \mathbf{D}_k^{(i)} \Sigma_k^{-1}\right]\right) + c_\Sigma \end{aligned} \quad (43)$$

According to Equation (35), $q^{(i+1)}(\Sigma_k)$ can be updated to an inverse Wishart probability density function with a degree-of-freedom parameter $\hat{u}_k^{(i+1)}$ and an inverse scale matrix $\hat{\mathbf{U}}_k^{(i+1)}$, which is represented as:

$$q^{(i+1)}(\Sigma_k) = \text{IW}\left(\Sigma_k; \hat{u}_k^{(i+1)}, \hat{\mathbf{U}}_k^{(i+1)}\right), \quad (44)$$

where the degree-of-freedom parameter $\hat{u}_k^{(i+1)}$ and an inverse scale matrix $\hat{\mathbf{U}}_k^{(i+1)}$ are represented as:

$$u_k^{(i+1)} = u_k + 1, \quad (45)$$

$$\mathbf{U}_k^{(i+1)} = \mathbf{U}_k + \mathbf{E}^{(i+1)}[\xi_k] \mathbf{D}_k^{(i)}. \quad (46)$$

Setting $\Phi = \mathbf{X}_k$ and bringing Equation (32) into equation (28), it can be obtained (Xu et al. 2013):

$$\begin{aligned} \log q^{(i+1)}(\mathbf{X}_k) &= -0.5E^{(i+1)}[\lambda_k](\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)^T \mathbf{R}_k^{-1} \\ &\quad \times (\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k) \\ &\quad - 0.5E^{(i+1)}[\xi_k](\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1})^T E^{(i+1)}[\Sigma_k^{-1}] \\ &\quad \times (\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1}) + c_{x_k} \end{aligned} \quad (47)$$

Define the modified one-step prediction probability density function $p^{(i+1)}(\mathbf{X}_{k|1:k-1})$ and the modified likelihood probability density function $p^{(i+1)}(\mathbf{Z}_{k|1} | \mathbf{X}_k)$ as:

$$p^{(i+1)}(\mathbf{X}_k | \mathbf{Z}_{1:k-1}) = N\left(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k-1}^{(i+1)}, \hat{\mathbf{P}}_{k|k-1}^{(i+1)}\right), \quad (48)$$

$$p^{(i+1)}(\mathbf{Z}_k | \mathbf{X}_k) = N\left(\mathbf{Z}_k; \mathbf{H}_k \mathbf{X}_k, \hat{\mathbf{R}}_k^{(i+1)}\right), \quad (49)$$

where the modified measurement noise covariance matrix $\hat{\mathbf{R}}_k^{(i+1)}$ and the modified prediction error covariance matrix $\hat{\mathbf{P}}_{k|k-1}^{(i+1)}$ are represented as:

$$\begin{cases} \hat{\mathbf{R}}_k^{(i+1)} = \frac{\mathbf{R}_k}{E^{(i+1)}[\lambda_k]} \\ \hat{\mathbf{P}}_{k|k-1}^{(i+1)} = \frac{\{E^{(i+1)}[\Sigma_k^{-1}]\}^{-1}}{E^{(i+1)}[\xi_k]} \end{cases} \quad (50)$$

According to Equations (48–50), it can be updated to a Gaussian probability density function with a mean vector $\hat{\mathbf{X}}_{k|k}^{(i+1)}$ and a covariance matrix $\mathbf{P}_{k|k}^{(i+1)}$, which is represented as:

$$q^{(i+1)}(\mathbf{X}_k) = \mathcal{N}\left(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k}^{(i+1)}, \mathbf{P}_{k|k}^{(i+1)}\right), \quad (51)$$

where

$$\hat{\mathbf{X}}_{k|k}^{(i+1)} = \mathbf{X}_{k|k-1} + \mathbf{K}_k^{(i+1)}(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k-1}), \quad (52)$$

$$\mathbf{P}_{k|k}^{(i+1)} = \mathbf{P}_{k|k-1}^{(i+1)} - \mathbf{K}_k^{(i+1)} \mathbf{H}_k \mathbf{P}_{k|k-1}^{(i+1)}, \quad (53)$$

where the modified $\mathbf{K}_k^{(i+1)}$ is represented as:

$$\mathbf{K}_k^{(i+1)} = \mathbf{P}_{k|k-1}^{(i+1)} \mathbf{H}_k^T \left(\mathbf{H}_k \mathbf{P}_{k|k-1}^{(i+1)} \mathbf{H}_k^T + \hat{\mathbf{R}}_k^{(i+1)} \right)^{-1}. \quad (54)$$

After N update iterations, the approximate a posteriori probability density functions $q(\mathbf{X}_k)$, $q(\xi_k)$, $q(\Sigma_k)$ and $q(\lambda_k)$ can be represented as:

$$\begin{aligned} q(\mathbf{X}_k) &\approx q^{(N)}(\mathbf{X}_k) = N\left(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k}^{(N)}, \mathbf{P}_{k|k}^{(N)}\right) \\ &= N\left(\mathbf{X}_k; \hat{\mathbf{X}}_{k|k}, \mathbf{P}_{k|k}\right), \end{aligned} \quad (55)$$

$$q(\xi_k) \approx q^{(N)}(\xi_k) = G\left(\xi_k; \alpha_k^N, \beta_k^N\right), \quad (56)$$

$$q(\Sigma_k) \approx q(\Sigma_k)^{(N)} = \text{IW}\left(\Sigma_k; \hat{u}_k^{(N)}, \hat{\mathbf{U}}_k^{(N)}\right), \quad (57)$$

$$q(\lambda_k) \approx q^{(N)}(\lambda_k) = G\left(\lambda_k; \delta_k^N, \gamma_k^N\right). \quad (58)$$

2. Calculation of expected value

The desired expectation values $E^{(i+1)}[\xi_k]$, $E^{(i+1)}[\lambda_k]$ and $E^{(i+1)}[\Sigma_k^{-1}]$ can be calculated according to Equations (35), (41) and (44) as follows:

$$\begin{cases} E^{(i+1)}[\xi_k] = \frac{\alpha_k^{i+1}}{\beta_k^{i+1}} \\ E^{(i+1)}[\lambda_k] = \frac{\delta_k^{i+1}}{\gamma_k^{i+1}} \\ E^{(i+1)}[\Sigma_k^{-1}] = \left(\hat{u}_k^{(i+1)} - n - 1\right) \left(\hat{\mathbf{U}}_k^{(i+1)}\right)^{-1} \end{cases} \quad (59)$$

According to Equation (51), the expected values $\mathbf{D}_k^{(i+1)}$ and $E_k^{(i+1)}$ are represented as:

$$\begin{aligned} \mathbf{D}_k^{(i+1)} &= E^{(i+1)}[(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1})(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1})^T] \\ &= E^{(i+1)}\left[(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k}^{(i+1)} + \hat{\mathbf{X}}_{k|k}^{(i+1)} - \hat{\mathbf{X}}_{k|k-1})\right. \\ &\quad \times \left.(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k}^{(i)} + \hat{\mathbf{X}}_{k|k}^{(i+1)} - \hat{\mathbf{X}}_{k|k-1})^T\right] \\ &= E^{(i+1)}\left[(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1}^{(i+1)})(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k-1}^{(i)})^T\right] + \\ &\quad (\hat{\mathbf{X}}_{k|k}^{(i)} - \hat{\mathbf{X}}_{k|k-1}^{(i+1)})(\hat{\mathbf{X}}_{k|k}^{(i+1)} - \hat{\mathbf{X}}_{k|k-1}^{(i)})^T \\ &= \mathbf{P}_{k|k}^{(i+1)} + (\hat{\mathbf{X}}_{k|k}^{(i+1)} - \hat{\mathbf{X}}_{k|k-1}^{(i+1)})(\hat{\mathbf{X}}_{k|k}^{(i+1)} - \hat{\mathbf{X}}_{k|k-1}^{(i+1)})^T \end{aligned} \quad (60)$$

TABLE 1 | Pseudo-code for the variational Bayesian extended Kalman filter.**Algorithm:** variational Bayesian extended Kalman filter**Inputs:** $\hat{\mathbf{X}}_{k-1|k-1}$, $\mathbf{P}_{k-1|k-1}$, $\mathbf{F}_{k-1}$, $\mathbf{H}_k$, $\mathbf{Z}_k$, $\mathbf{Q}_{k-1}$, $\mathbf{R}_k$, m , n , ω , ν , τ , N **Time update:**

1. Based on equations (8) and (9), calculate $\mathbf{X}_{k|k-1}$ and $\mathbf{P}_{k|k-1}$

Measurement update:

2. Initialization: $u_k = n + \tau + 1$, $\mathbf{U}_k = \tau \mathbf{P}_{k|k-1}$, $\hat{\mathbf{X}}_{k|k}^{(0)} = \hat{\mathbf{X}}_{k|k-1}$, $\mathbf{P}_{k|k}^{(0)} = \mathbf{P}_{k|k-1}$, $E^{(0)}[\Sigma_k^{-1}] = (u_k - n - 1)\mathbf{U}_k^{-1}$

for $i = 0: N - 1$

4. Calculate $\mathbf{D}_k^{(i)}$, α_k^{i+1} and β_k^{i+1} based on equations (34), (36), (37) and (60) and update $q^{(i+1)}(\xi_k)$ based on equation (30)
5. Calculate $E_k^{(i)}$, γ_k^{i+1} and δ_k^{i+1} based on equations (39), (41), (42) and (61) and update $q^{(i+1)}(\lambda_k)$ based on equation (35)
6. Calculate $u_k^{(i+1)}$ and $\mathbf{U}_k^{(i+1)}$ based on equations (45) and (42) and update $q^{(i+1)}(\Sigma_k)$ based on equation (44)
7. Calculate $\tilde{\mathbf{R}}_k^{(i+1)}$ and $\tilde{\mathbf{P}}_{k|k-1}^{(i+1)}$ based on equation (50) and update $q^{(i+1)}(\mathbf{X}_k)$ based on equations (51)–(64)

end

8. $\hat{\mathbf{X}}_{k|k} = \hat{\mathbf{X}}_{k|k}^{(N)}$, $\mathbf{P}_{k|k} = \mathbf{P}_{k|k}^{(N)}$

Outputs: $\hat{\mathbf{X}}_{k|k}$ and $\mathbf{P}_{k|k}$

$$\begin{aligned}
 E_k^{(i+1)} &= E^{(i+1)}[(\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)(\mathbf{Z}_k - \mathbf{H}_k \mathbf{X}_k)^T] \\
 &= E^{(i+1)}\left[\left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)} + \mathbf{H}_k (\hat{\mathbf{X}}_{k|k}^{(i+1)} - \mathbf{X}_{k|k})\right)\right. \\
 &\quad \left. \times \left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)} + \mathbf{H}_k (\hat{\mathbf{X}}_{k|k}^{(i+1)} - \mathbf{X}_{k|k})\right)^T\right] \\
 &= \left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)}\right) \left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)}\right)^T + \mathbf{H}_k E^{(i+1)} \\
 &\quad \left[\left(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k}^{(i+1)}\right) \left(\mathbf{X}_k - \hat{\mathbf{X}}_{k|k}^{(i+1)}\right)^T\right] \mathbf{H}_k^T \\
 &= \left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)}\right) \left(\mathbf{Z}_k - \mathbf{H}_k \hat{\mathbf{X}}_{k|k}^{(i+1)}\right)^T + \mathbf{H}_k \mathbf{P}_{k|k}^{(i+1)} \mathbf{H}_k^T
 \end{aligned} \quad (61)$$

The computational flow of the VBEKF algorithm is depicted in Table 1.

2.3 | Path Tracking Controller Design

As depicted in Figure 1, the position information in the global coordinate system XOY is $P = [X_O, Y_O, \theta_O]^T$ and the velocity information is $q = [v, \omega]^T$, and the position information and velocity information at the point P_r of the reference path are $P_r = [X_r, Y_r, \theta_r]^T$ and $q_r = [v_r, \omega_r]^T$, respectively. Therefore, the tracking error of the tracked robot in the local coordinate system xoy is obtained as $p_e = [x_e, y_e, \theta_e]^T$, which can be represented as (Sabiha et al. 2022):

$$\begin{aligned}
 \begin{bmatrix} x_e \\ y_e \\ \theta_e \end{bmatrix} &= \begin{bmatrix} \cos \theta + \sin \theta \tan \phi & \sin \theta - \cos \theta \tan \phi & 0 \\ -\sin \theta + \cos \theta \tan \phi & \cos \theta + \sin \theta \tan \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X_r - X_O \\ Y_r - Y_O \\ \theta_r - \theta_O \end{bmatrix} \quad (62)
 \end{aligned}$$

Combining Equations (2) and (62) gives the differential equation for the path tracking error of the tracked robot, which is represented as:

$$\begin{bmatrix} \dot{x}_e \\ \dot{y}_e \\ \dot{\theta}_e \end{bmatrix} = \begin{bmatrix} \omega y_e - v + v_r \cos \theta_e \\ -\omega x_e + v \sin \theta_e \\ \dot{\theta}_r - \omega \end{bmatrix}. \quad (63)$$

Combined with Equation (62) the following Lyapunov function can be designed as:

$$V = \frac{1}{2} \left(x_e^2 + y_e^2 \right) + \frac{1}{k_2} \left(1 - \cos \theta_e \right). \quad (64)$$

The first order derivative of V can be represented as:

$$\begin{aligned}
 \dot{V} &= x_e \dot{x}_e + y_e \dot{y}_e + \frac{1}{k_2} \dot{\theta}_e \sin \theta_e \\
 &= x_e (\omega y_e - v + v_r \cos \theta_e) \\
 &\quad + y_e (-\omega x_e + v \sin \theta_e) + \frac{1}{k_2} (\omega_r - \omega) \sin \theta_e \\
 &= -k_1 x_e^2 - \frac{k_3}{k_2} v_r \sin^2 \theta_e
 \end{aligned} \quad (65)$$

The auxiliary control input is selected as (Sabiha et al. 2022):

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} v_r \cos \theta_e + k_1 x_e \\ \omega_r + k_2 v_r y_e + k_3 v_r \sin \theta_e \end{bmatrix}. \quad (66)$$

The actual control volume is represented as (Lu et al. 2023b):

$$\begin{cases} v_L = \frac{v}{1 - i_L} - \frac{b\omega}{2(1 - i_L)} \\ v_R = \frac{v}{1 - i_R} - \frac{b\omega}{2(1 - i_R)} \end{cases}. \quad (67)$$

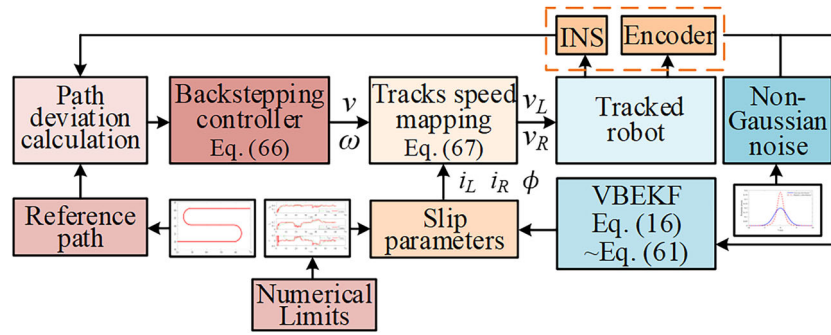
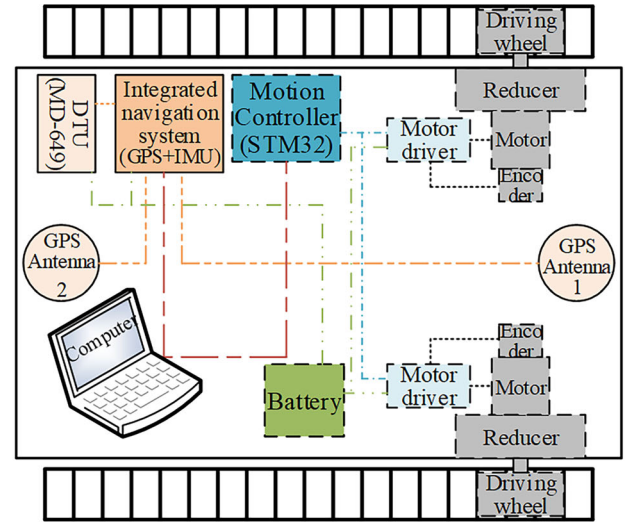


FIGURE 4 | Structural diagram of the slip compensation backstepping controller for the tracked robot.



(a)



(b)

FIGURE 5 | Tracked robot system: (a) tracked robot and (b) control and data acquisition system.

TABLE 2 | Structural parameters of the tracked robot.

Parameters	Values
Length $\times$ width $\times$ height ($L \times W \times H$)/mm	1400 $\times$ 900 $\times$ 520
Mass M /kg	121.58
Tracks' center distance b /mm	750
Length of track grounding L_g /mm	828
Width of track grounding W_g /mm	150
Reduction ratio r_i	20
Radius of driving wheel r /mm	150.925

Bringing Equation (66) into Equation (65), it can be obtained:

$$\dot{V} = -k_1 x_e^2 - \frac{k_3}{k_2} v_r \sin^2 \theta_e, \quad (68)$$

where when $k_1 > 0$, $k_2 > 0$, $k_3 > 0$ and $v_r > 0$, it can be obtained that $\dot{V} \leq 0$. So the selected control inputs satisfy the Lyapunov

stability criterion, and it can be proved that the system is globally asymptotically stable (Figure 4).

3 | Results and Analysis

3.1 | Simulation and Experimental Conditions

To validate the slip parameters estimation and path tracking control method proposed in this paper, a tracked robot system is constructed as depicted in Figure 5, and its structural parameters are depicted in Table 2 (Zhao et al. 2024).

The tracked robot is equipped with MS-6111 integrated navigation system, which consists of a high-precision global navigation satellite system (GNSS) module and an inertial measurement unit (IMU). This system offers an RTK differential positioning accuracy of 1 cm, a heading accuracy of 0.1° , a horizontal velocity accuracy of 0.1 m/s, an accelerometer sensitivity of 0.1 mg, and a sampling frequency of 10 Hz. In the field experiment, path planning is accomplished on the upper computer, followed by the acquisition of the tracked robot's position information by the MCU through serial communication.

3.2 | Simulation Analysis

3.2.1 | Slip Parameter Estimation Simulations

In real field environments, external noises are often non-Gaussian noises with heavy-tailed characteristics, so it cannot be directly modeled using a Gaussian distribution. To better simulate the complex environment in the field, non-Gaussian noise with heavy-tailed characteristics is established through Equations (69) and (70) (Huang et al. 2017; Chen et al. 2021; Xu et al. 2019):

$$W_k \sim \begin{cases} N(0, Q_w) & w.p. \quad 1 - P_w \\ N(0, M_w Q_w) & w.p. \quad P_w \end{cases}, \quad (69)$$

$$V_k \sim \begin{cases} N(0, R_v) & w.p. \quad 1 - P_v \\ N(0, M_v R_v) & w.p. \quad P_v \end{cases}, \quad (70)$$

where Q_w and R_v represent the covariance matrices of process noise and observation noise, M_w and M_v represent the covariance matrix amplification factors, and *w.p.* represents the probability, and P_w and P_v represent the probability of the existence of outliers in the process noise and observation noise.

In the numerical simulations, non-Gaussian noise is added to the track speed of the tracked robot as depicted in Figure 6a, and the slip parameters between the track and the track are depicted in Figure 6b.

Based on the simulation conditions shown in Figure 6, the VBEKF algorithm proposed in this paper is compared with the traditional EKF algorithm in references (Zhao et al. 2024; Ha et al. 2013). And the traveling paths of the tracked robot are predicted by combining the left and right track velocities of the tracked robot and the slip kinematics model. As depicted in Figure 7a, compared with the EKF algorithm, the VBEKF algorithm is more resistant to interference, and can estimate the slip parameters more stably and accurately. As depicted in Figure 7b, the VBEKF algorithm estimates the slip parameters having a smaller error. As depicted in Figure 7c,d, it can be seen

that the path and heading angle predicted by the slip parameters estimated by the VBEKF algorithm are closer to the real path and heading angle. As depicted in Figure 7e, it can be seen that the path and heading angle predicted by the slip parameters estimated using the VBEKF algorithm exhibit a smaller error compared to EKF algorithm. From the above results, it can be seen that VBEKF predicts the slip parameters more accurately than EKF in a non-Gaussian noise field environment. The errors in predicting slip parameters (i_L , i_R , ϕ) using the EKF and VBEKF algorithms are depicted in Table 3. It can be seen that the proposed VBEKF algorithm greatly reduces the mean absolute error, maximum error and standard deviation of slip parameter estimation. In summary, compared with the traditional EKF algorithm in references (Zhao et al. 2024; Ha et al. 2013), the VBEKF algorithm proposed in this paper has higher accuracy in estimating slip parameters during the driving process of tracked robot, which proves the effectiveness of the proposed method.

3.2.2 | Path Tracking Control Simulations

According to the previous analysis, the slip parameters estimated by the VBEKF algorithm proposed in this paper are more accurate than the traditional EKF algorithm (Zhao et al. 2024; Ha et al. 2013). Therefore, in the process of path tracking control for tracked robots, the VBEKF algorithm is employed to accurately estimate the slip parameters, which are subsequently utilized in the path tracking control system for effective slip compensation. This approach is contrasted with the backstepping controller presented in reference (Sabiha et al. 2022), which does not incorporate slip compensation.

The simulation results of path tracking control for the tracked robot are depicted in Figure 8. As depicted in Figure 8a, it can be seen that the path tracking accuracy with slip parameters compensation is significantly higher than that without slip parameters compensation. As depicted in Figure 8b, it can be seen that the inputs of the controller with slip parameters compensation are larger. As depicted in Figure 8c, it can be seen

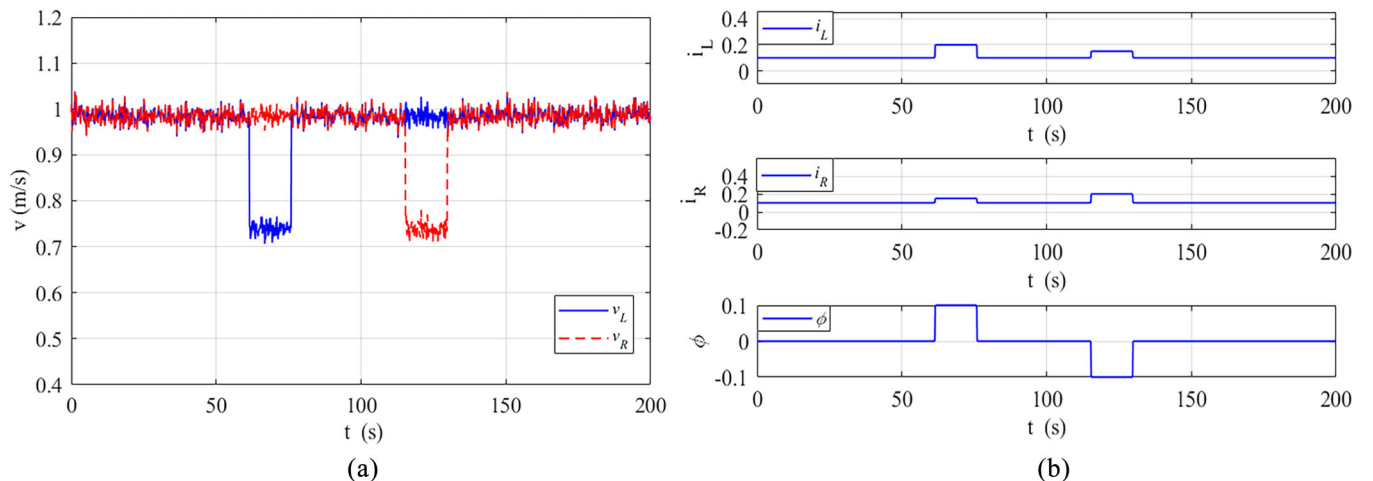


FIGURE 6 | Simulated track speed and slip parameters of the tracked robot: (a) speeds of track and (b) slip parameters.

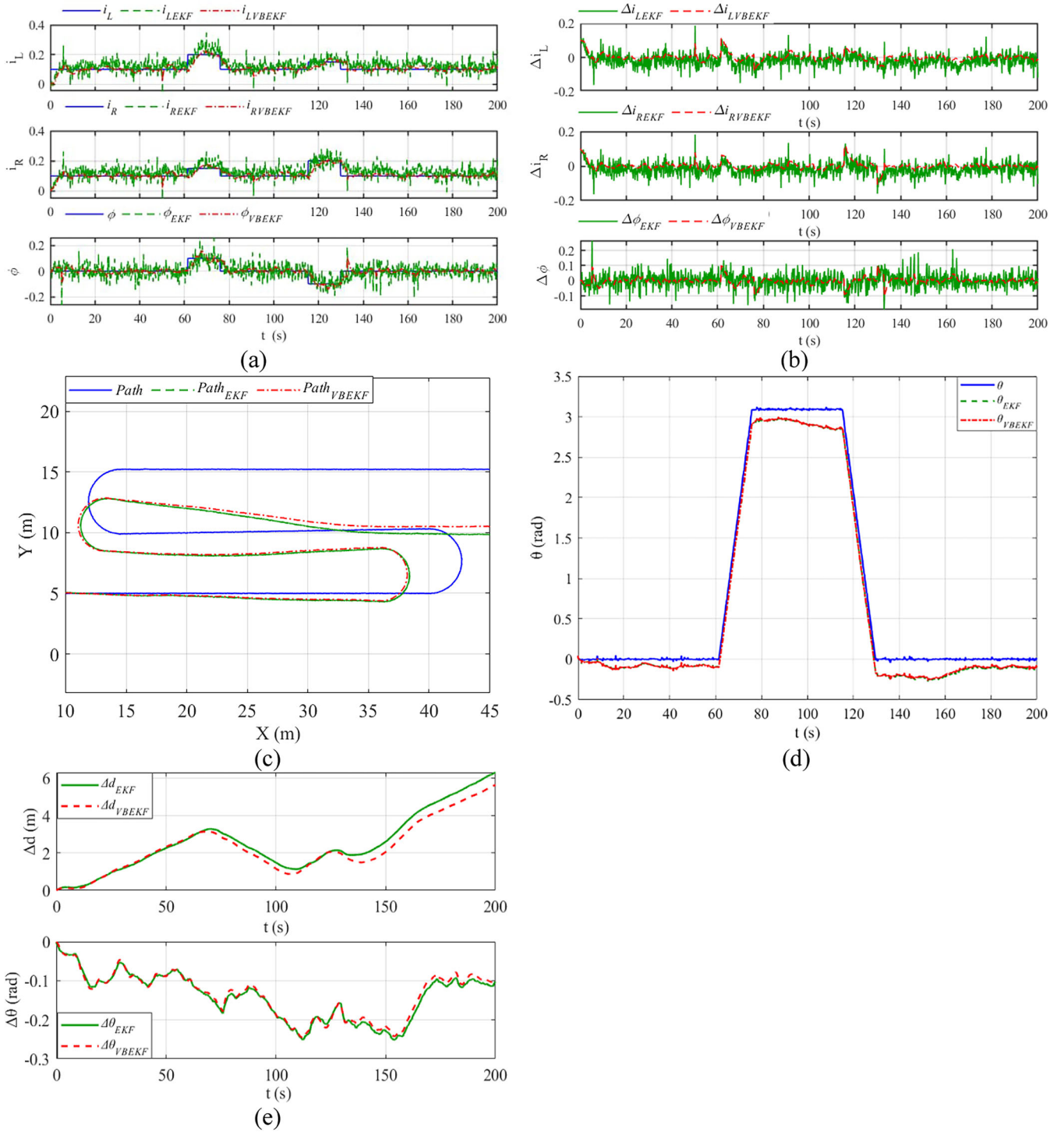


FIGURE 7 | Simulation results of slip parameter estimation under the simulated field environment: (a) slip parameter estimation; (b) estimation errors; (c) path prediction; (d) heading angle prediction; and (e) path and heading angle errors.

that the tracking errors of the controller with slip compensation are significantly smaller, with a notable reduction of 25% in the maximum lateral error compared to the uncompensated case. As depicted in Figure 8d, it can be seen that the cumulative errors of the controller with slip parameters compensation are much smaller. Under the same tracking path length, the cumulative errors are reduced by 20.6% after slip parameters compensation.

3.3 | Experimental Verification

3.3.1 | Slip Parameters Estimation Experiments

Based on the position information of the tracked robot and the velocity information of the left and right tracks collected from the experiments, the results of the slip parameters estimation are depicted in Figure 9a, it can be seen that when the tracked robot is

TABLE 3 | Statistics of slip parameter estimation errors under field environment.

Slip parameters	Method	Mean absolute error	Maximum error	Standard deviation
i_L	EKF	0.0332	0.1863	0.0259
	VBEKF	0.0151	0.1135	0.0175
i_R	EKF	0.0328	0.1823	0.0254
	VBEKF	0.0132	0.1210	0.0177
ϕ	EKF	0.0369	0.2675	0.0313
	VBEKF	0.0153	0.1068	0.0177

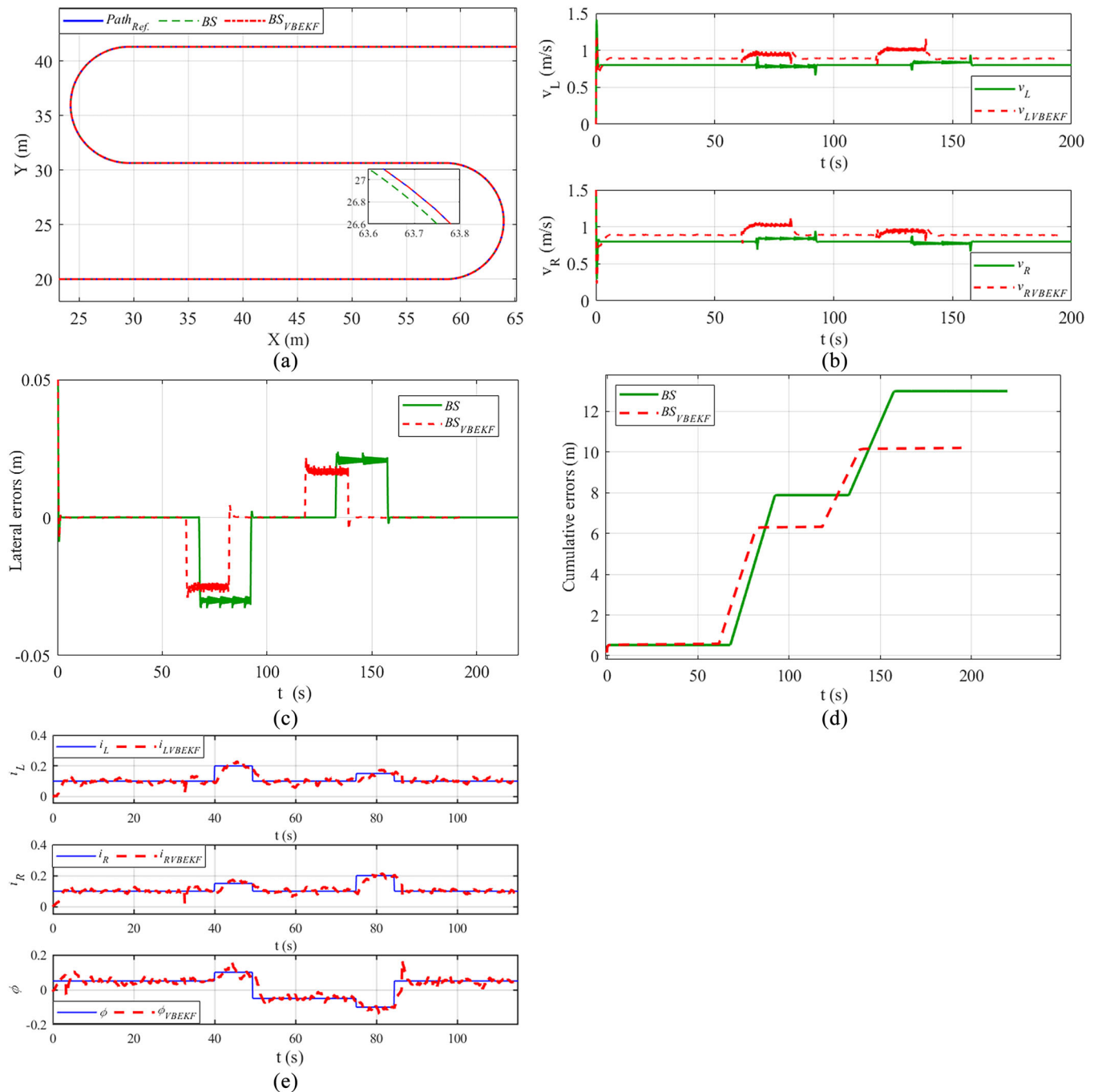


FIGURE 8 | Simulation results of path tracking control for the tracked robot: (a) path comparison; (b) control inputs; (c) lateral errors; (d) cumulative lateral errors; and (e) slip parameters.

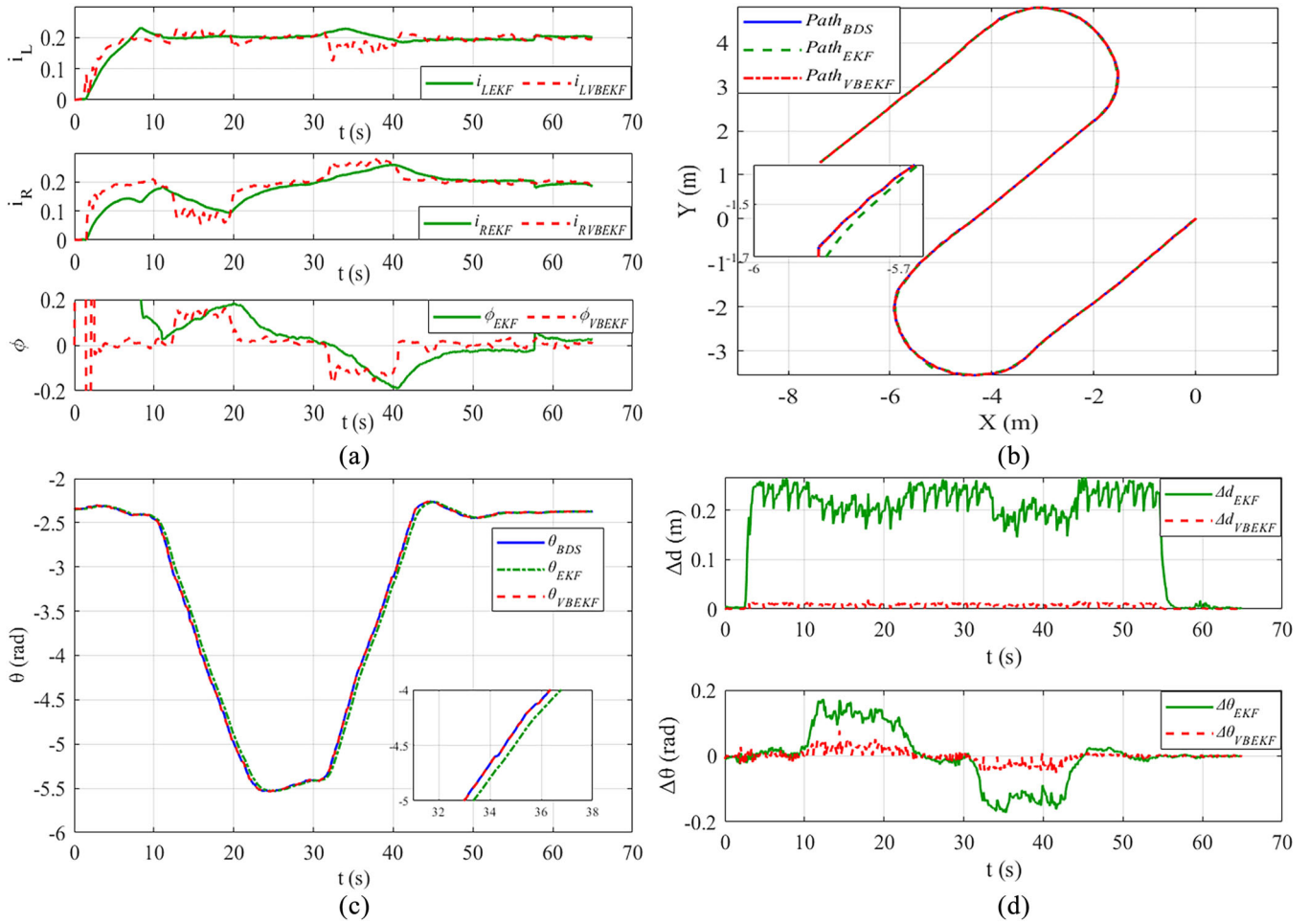


FIGURE 9 | Experimental results of slip parameter estimation: (a) slip parameters; (b) path prediction; (c) heading angle prediction; and (d) path and heading angle prediction errors.

traveling in a straight line, the slip parameters estimated by the EKF and VBEKF algorithms are relatively close to each other. However, in the case of the tracked robot turning a corner, the VBEKF algorithm responds faster, and the slip parameters estimated by the VBEKF algorithm are more in line with the actual situation, and the i_L , i_R , and ϕ estimated in the case of turning all show the expected changes. The predicted traveling path and heading angle based on the speeds of the left and right tracks and the slip kinematics model are depicted in Figure 9b,c, respectively, and it can be seen that the path and the heading angle predicted based on the slip parameters estimated by the VBEKF algorithm are more accurate. The predicted path errors and heading angle errors are depicted in Figure 9d, from which it can be seen that the path and heading angle errors predicted by the slip parameters estimated by the VBEKF algorithm are significantly smaller. Based on the experimental results above, the VBEKF algorithm outperforms the traditional EKF algorithm in estimating slip parameters, as compared to the traditional EKF algorithm in references (Zhao et al. 2024; Ha et al. 2013).

3.3.2 | Path Tracking Control Experiments

The experimental results of the path tracking control of the tracked robot based on the slip parameters compensation are depicted in

Figure 10. The slip parameters estimated by VBEKF are depicted in Figure 10a. The comparison of path tracking performance is depicted in Figure 10b, where it is evident that the tracking accuracy of the backstepping controller with slip parameter compensation is significantly higher, resulting in reduced errors particularly during the turning maneuvers of the tracked robot, compared to the controller without slip compensation as described in reference (Wang et al. 2021). The control inputs of tracked robot are depicted in Figure 10c. The lateral errors are depicted in Figure 10d, from which it can be seen that the tracking errors with slip compensation are significantly smaller. Specifically, the maximum lateral error has been reduced from 25 to 10 cm, indicating a substantial improvement in tracking accuracy. Furthermore, the mean absolute error has also undergone a notable decrease, dropping from 8.5 to 3.5 cm. Based on the experimental results above, the backstepping controller designed in this paper can achieve better path tracking control effect after combining the slip parameters estimated by the VBEKF algorithm, which verifies the simulation results mentioned above.

4 | Conclusions

In this paper, a slip compensation-based path tracking control method for tracked robots is proposed. First, the noise is

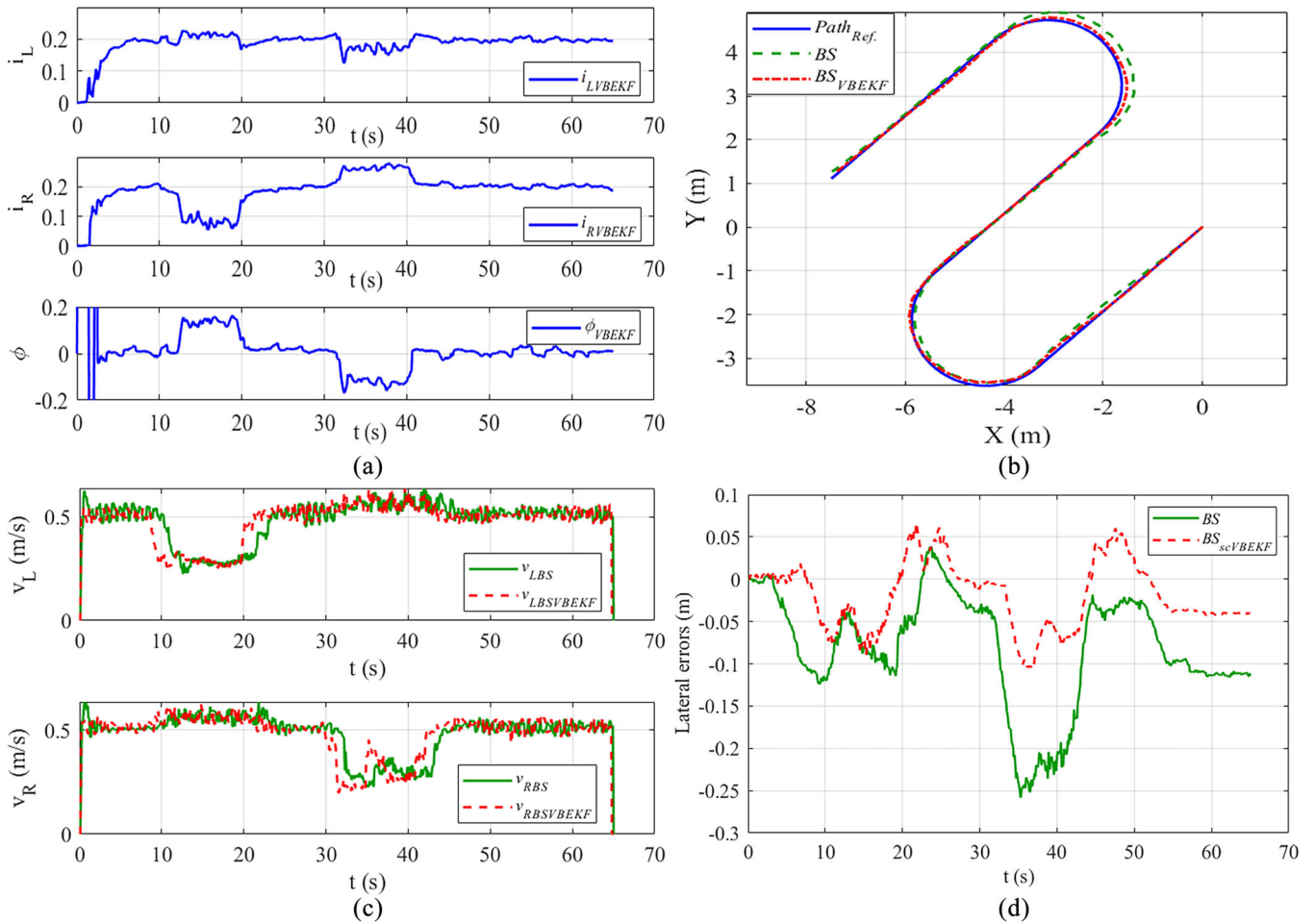


FIGURE 10 | Experimental results of path tracking control for the tracked robot: (a) slip parameter estimation; (b) path tracking; (c) control inputs; and (d) lateral errors.

modeled non-Gaussian to establish noise closer to the field environment. Based on the established slip kinematics model of a tracked robot, a VBEKF algorithm is proposed to estimate the slip parameters between the tracks and the ground. On this basis, a path tracking controller with slip parameters compensation is designed, and the slip parameters estimated by VBEKF are utilized for path tracking control compensation. Finally, it is verified by numerical simulations and experiments.

The simulation and experimental results show that, compared with the traditional EKF algorithm, the VBEKF algorithm proposed in this paper is characterized by higher accuracy and better robustness in the complex field environment. The mean error, the maximum error, and the standard deviation of the estimated slip parameters have all significantly decreased. After compensating the estimated slip parameters by VBEKF algorithm to the backstepping-based path tracking controller, the mean absolute lateral error is reduced from 8.5 to 2.5 cm under the same tracking path length, which greatly improves the path tracking accuracy. Moreover, the VBEKF algorithm proposed in this paper does not need soil parameters to estimate the slip parameters, so it can be used in a variety of field operating environments. The research in this paper provides new ideas and theoretical guidance for slip parameter estimation and path tracking control of tracked robots. In our future

work, we will further consider the control strategies of tracked robots in situations such as obstacle avoidance and multi-machine collaboration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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